

Lecture 5: Sorting Lower Bound and “Linear-Time” Sorting

Jessica Sorrell

September 15, 2026

601.433/633 Introduction to Algorithms

Slides by Mike Dinitz

Introduction

Lots of ways of sorting in $O(n \log n)$ time: mergesort, heapsort, randomized quicksort, deterministic quicksort with BFPRT pivot selection, ...

Is it possible to do better?

Introduction

Lots of ways of sorting in $O(n \log n)$ time: mergesort, heapsort, randomized quicksort, deterministic quicksort with BFPRT pivot selection, ...

Is it possible to do better? No!

Introduction

Lots of ways of sorting in $O(n \log n)$ time: mergesort, heapsort, randomized quicksort, deterministic quicksort with BFPRT pivot selection, ...

Is it possible to do better? No! And yes!

Introduction

Lots of ways of sorting in $O(n \log n)$ time: mergesort, heapsort, randomized quicksort, deterministic quicksort with BFPRT pivot selection, ...

Is it possible to do better? No! And yes!

Comparison Model: we are given a constant-time algorithm which can compare any two elements. No other information about elements.

- ▶ All algorithms we've seen so far have been in this model

Introduction

Lots of ways of sorting in $O(n \log n)$ time: mergesort, heapsort, randomized quicksort, deterministic quicksort with BFPRT pivot selection, ...

Is it possible to do better? No! And yes!

Comparison Model: we are given a constant-time algorithm which can compare any two elements. No other information about elements.

- ▶ All algorithms we've seen so far have been in this model

No: every algorithm in the comparison model must have worst-case running time $\Omega(n \log n)$.

Yes: If we assume extra structure for the elements, can do sorting in $O(n)$ time*

Sorting Lower Bound

Statement

Theorem

Any sorting algorithm in the comparison model must make at least $\log(n!) = \Theta(n \log n)$ comparisons (in the worst case).

Lower bound on the number of comparisons – running time could be even worse!
Allows algorithm to reorder elements, copy them, move them, etc. for free.

Statement

Theorem

Any sorting algorithm in the comparison model must make at least $\log(n!) = \Theta(n \log n)$ comparisons (in the worst case).

Lower bound on the number of comparisons – running time could be even worse!

Allows algorithm to reorder elements, copy them, move them, etc. for free.

Why is this hard?

- ▶ Lower bound needs to hold for *all* algorithms
- ▶ How can we simultaneously reason about algorithms as different as mergesort, quicksort, heapsort, ...?

Sorting as Permutations

Think of an array $\mathbf{A}$ as a *permutation*: $\mathbf{A}[i]$ is the $\pi(i)$ 'th smallest element

$$\mathbf{A} = [23, 14, 2, 5, 76]$$

Corresponds to $\pi = (3, 2, 0, 1, 4)$:

$$\mathbf{A}' = [2, 5, 14, 23, 76]$$

$$\pi(0) = 3$$

$$\pi(1) = 2$$

$$\pi(2) = 0$$

$$\pi(3) = 1$$

$$\pi(4) = 4$$

Sorting as Permutations

Think of an array $\mathbf{A}$ as a *permutation*: $\mathbf{A}[i]$ is the $\pi(i)$ 'th smallest element

$$\mathbf{A} = [23, 14, 2, 5, 76]$$

Corresponds to $\pi = (3, 2, 0, 1, 4)$:

$$\mathbf{A}' = [2, 5, 14, 23, 76]$$

$$\pi(0) = 3 \qquad \pi(1) = 2 \qquad \pi(2) = 0 \qquad \pi(3) = 1 \qquad \pi(4) = 4$$

In other words, $\pi(i)$ tells us what index in the sorted array the i 'th element of the original array ends up in.

Sorting as Permutations

Think of an array $\mathbf{A}$ as a *permutation*: $\mathbf{A}[i]$ is the $\pi(i)$ 'th smallest element

$$\mathbf{A} = [23, 14, 2, 5, 76]$$

Corresponds to $\pi = (3, 2, 0, 1, 4)$:

$$\mathbf{A}' = [2, 5, 14, 23, 76]$$

$$\pi(0) = 3 \qquad \pi(1) = 2 \qquad \pi(2) = 0 \qquad \pi(3) = 1 \qquad \pi(4) = 4$$

In other words, $\pi(i)$ tells us what index in the sorted array the i 'th element of the original array ends up in.

Lemma

Given $\mathbf{A}$ with $|\mathbf{A}| = n$, if we can sort in $T(n)$ comparisons then we can find π in $T(n)$ comparisons.

Sorting As Permutations (cont'd)

Lemma

Given $\mathbf{A}$ with $|\mathbf{A}| = n$, if we can sort in $T(n)$ comparisons then we can find π in $T(n)$ comparisons.

Proof Sketch.

- ▶ “Tag” each element of $\mathbf{A}$ with index:
 $[23, 14, 2, 5, 76] \rightarrow [(23, 0), (14, 1), (2, 2), (5, 3), (76, 4)]$
- ▶ Sort tagged $\mathbf{A}$ into tagged $\mathbf{B}$ with $T(n)$ comparisons:
 $[(2, 2), (5, 3), (14, 1), (23, 0), (76, 4)]$
- ▶ Iterate through to get π : $\pi(2) = 0, \pi(3) = 1, \pi(1) = 2, \pi(0) = 3, \pi(4) = 4$



Sorting As Permutations (cont'd)

Lemma

Given $\mathbf{A}$ with $|\mathbf{A}| = n$, if we can sort in $T(n)$ comparisons then we can find π in $T(n)$ comparisons.

Proof Sketch.

- ▶ “Tag” each element of $\mathbf{A}$ with index:
 $[23, 14, 2, 5, 76] \rightarrow [(23, 0), (14, 1), (2, 2), (5, 3), (76, 4)]$
- ▶ Sort tagged $\mathbf{A}$ into tagged $\mathbf{B}$ with $T(n)$ comparisons:
 $[(2, 2), (5, 3), (14, 1), (23, 0), (76, 4)]$
- ▶ Iterate through to get π : $\pi(2) = 0, \pi(3) = 1, \pi(1) = 2, \pi(0) = 3, \pi(4) = 4$ □

Corollary

If we need at least $T(n)$ comparisons to find π , then we need at least $T(n)$ comparisons to sort!

Generic Algorithm

Want to show that it takes $\Omega(n \log n)$ comparisons to find π in comparison model.

- ▶ Only comparisons cost us anything!

Generic Algorithm

Want to show that it takes $\Omega(n \log n)$ comparisons to find π in comparison model.

- ▶ Only comparisons cost us anything!

Arbitrary algorithm:

- ▶ Starts with some comparison (e.g., compares $A[0]$ to $A[1]$)
- ▶ Rules out some possible permutations!
 - ▶ If $A[0] < A[1]$ then $\pi(0) < \pi(1)$
 - ▶ If $A[0] > A[1]$ then $\pi(0) > \pi(1)$
- ▶ Depending on outcome, choose next comparison to make.
- ▶ Continue until only one possible permutation.

Generic Algorithm

Want to show that it takes $\Omega(n \log n)$ comparisons to find π in comparison model.

- ▶ Only comparisons cost us anything!

Arbitrary algorithm:

- ▶ Starts with some comparison (e.g., compares $A[0]$ to $A[1]$)
- ▶ Rules out some possible permutations!
 - ▶ If $A[0] < A[1]$ then $\pi(0) < \pi(1)$
 - ▶ If $A[0] > A[1]$ then $\pi(0) > \pi(1)$
- ▶ Depending on outcome, choose next comparison to make.
- ▶ Continue until only one possible permutation.

Remind you of anything?

Decision Trees

Model any algorithm as a *binary decision tree*

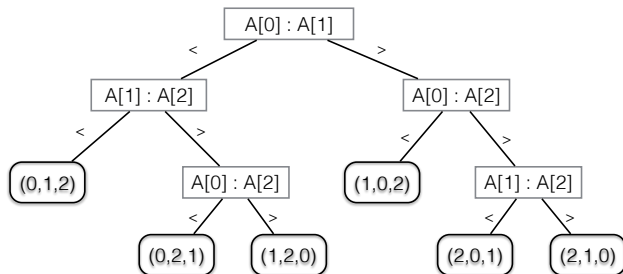
- ▶ Internal nodes: comparisons
- ▶ Leaves: permutations

Decision Trees

Model any algorithm as a *binary decision tree*

- ▶ Internal nodes: comparisons
- ▶ Leaves: permutations

Example: $n = 3$. Six possible permutations.

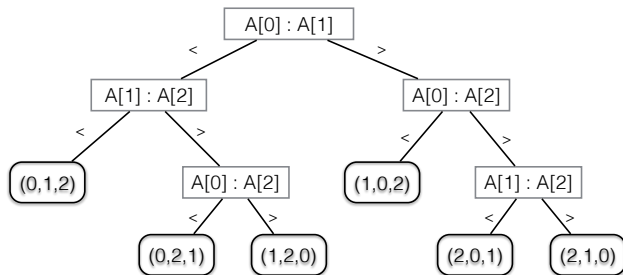


Decision Trees

Model any algorithm as a *binary decision tree*

- ▶ Internal nodes: comparisons
- ▶ Leaves: permutations

Example: $n = 3$. Six possible permutations.



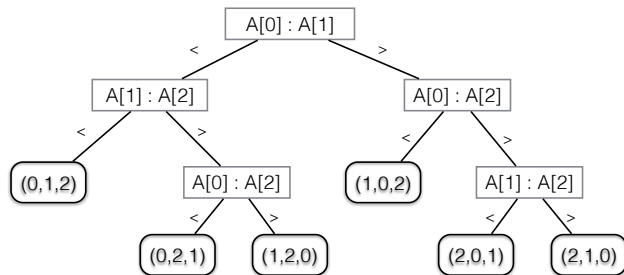
Max # comparisons:

Decision Trees

Model any algorithm as a *binary decision tree*

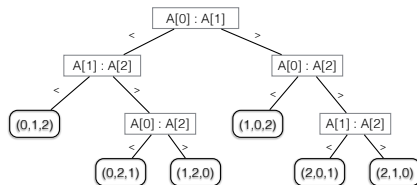
- ▶ Internal nodes: comparisons
- ▶ Leaves: permutations

Example: $n = 3$. Six possible permutations.



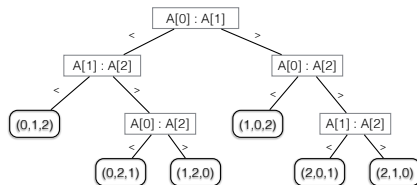
Max # comparisons: 3

Finishing Up



Scale to general n . Consider arbitrary decision tree.

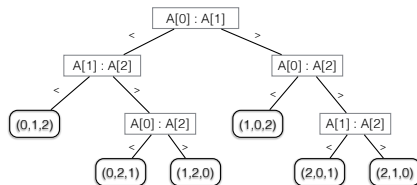
Finishing Up



Scale to general n . Consider arbitrary decision tree.

Max # comparisons

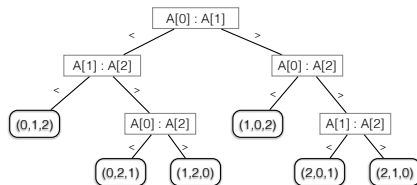
Finishing Up



Scale to general n . Consider arbitrary decision tree.

Max # comparisons = depth of tree

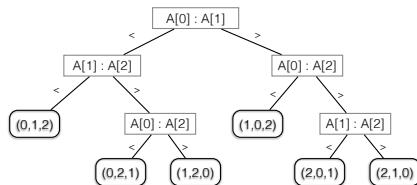
Finishing Up



Scale to general n . Consider arbitrary decision tree.

Max # comparisons = depth of tree
 $\geq \log_2(\# \text{ leaves})$

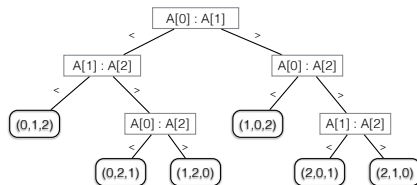
Finishing Up



Scale to general n . Consider arbitrary decision tree.

$$\begin{aligned}\text{Max } \# \text{ comparisons} &= \text{depth of tree} \\ &\geq \log_2(\# \text{ leaves}) \\ &= \log_2(n!)\end{aligned}$$

Finishing Up



Scale to general n . Consider arbitrary decision tree.

$$\begin{aligned}\text{Max } \# \text{ comparisons} &= \text{depth of tree} \\ &\geq \log_2(\# \text{ leaves}) \\ &= \log_2(n!) \\ &= \Theta(n \log n)\end{aligned}$$

Sorting Lower Bound Summary

Theorem

Every sorting algorithm in the comparison model must make at least $\log(n!) = \Theta(n \log n)$ comparisons (in the worst case).

Proof Sketch.

1. Lower bound on finding permutation $\pi \implies$ lower bound on sorting
 2. Any algorithm for finding π is a binary decision tree with $n!$ leaves.
 3. Any binary decision tree with $n!$ leaves has depth $\geq \log(n!) = \Theta(n \log n)$
- $\implies$ Every algorithm has worst case number of comparisons at least $\Theta(n \log n)$. □

“Linear-Time” Sorting

Bypassing the Lower Bound

What if we're *not* in the comparison model?

- ▶ Can do more than just compare elements.

Main example: *integers*.

- ▶ What is the **3rd** bit of $A[0]$?
- ▶ Is $A[0] \ll k$ larger than $A[1] \gg c$?
- ▶ Is $A[0]$ even?

Same ideas apply to letters, strings, etc.

Counting Sort

Suppose ***A*** consists of ***n*** integers, all in $\{0, 1, \dots, k - 1\}$.

Counting Sort

Suppose $\mathbf{A}$ consists of n integers, all in $\{0, 1, \dots, k - 1\}$.

Counting Sort:

- ▶ Maintain an array $\mathbf{B}$ of length k initialized to all 0
- ▶ Scan through $\mathbf{A}$ and increment $\mathbf{B}[\mathbf{A}[i]]$.
- ▶ Scan through $\mathbf{B}$, output i exactly $\mathbf{B}[i]$ times.

Counting Sort

Suppose $\mathbf{A}$ consists of n integers, all in $\{0, 1, \dots, k - 1\}$.

Counting Sort:

- ▶ Maintain an array $\mathbf{B}$ of length k initialized to all 0
- ▶ Scan through $\mathbf{A}$ and increment $\mathbf{B}[\mathbf{A}[i]]$.
- ▶ Scan through $\mathbf{B}$, output i exactly $\mathbf{B}[i]$ times.

Correctness: Obvious

Counting Sort

Suppose $\mathbf{A}$ consists of n integers, all in $\{0, 1, \dots, k - 1\}$.

Counting Sort:

- ▶ Maintain an array $\mathbf{B}$ of length k initialized to all 0
- ▶ Scan through $\mathbf{A}$ and increment $\mathbf{B}[\mathbf{A}[i]]$.
- ▶ Scan through $\mathbf{B}$, output i exactly $\mathbf{B}[i]$ times.

Correctness: Obvious

Running time:

Counting Sort

Suppose $\mathbf{A}$ consists of n integers, all in $\{0, 1, \dots, k - 1\}$.

Counting Sort:

- ▶ Maintain an array $\mathbf{B}$ of length k initialized to all 0
- ▶ Scan through $\mathbf{A}$ and increment $\mathbf{B}[\mathbf{A}[i]]$.
- ▶ Scan through $\mathbf{B}$, output i exactly $\mathbf{B}[i]$ times.

Correctness: Obvious

Running time: $O(n + k)$

Bucket Sort: Counting Sort++

Often want to sort *objects* based on *keys*:

- ▶ Each object has a key: integer in $\{0, 1, \dots, k - 1\}$
- ▶ **A** consists of **n** objects

Bucket Sort: Counting Sort++

Often want to sort *objects* based on *keys*:

- ▶ Each object has a key: integer in $\{0, 1, \dots, k-1\}$
- ▶ **A** consists of **n** objects

Bucket Sort:

- ▶ Same idea as counting sort, but **B**[*i*] is bucket of objects with key *i*
- ▶ Bucket is a linked list with pointers to beginning and end
- ▶ Insert at *end* of list, using end pointer.
- ▶ For output, go through each bucket in order.

Bucket Sort: Counting Sort++

Often want to sort *objects* based on *keys*:

- ▶ Each object has a key: integer in $\{0, 1, \dots, k-1\}$
- ▶ **A** consists of **n** objects

Bucket Sort:

- ▶ Same idea as counting sort, but **B**[*i*] is bucket of objects with key *i*
- ▶ Bucket is a linked list with pointers to beginning and end
- ▶ Insert at *end* of list, using end pointer.
- ▶ For output, go through each bucket in order.

Running time:

Bucket Sort: Counting Sort++

Often want to sort *objects* based on *keys*:

- ▶ Each object has a key: integer in $\{0, 1, \dots, k - 1\}$
- ▶ **A** consists of **n** objects

Bucket Sort:

- ▶ Same idea as counting sort, but **B**[*i*] is bucket of objects with key *i*
- ▶ Bucket is a linked list with pointers to beginning and end
- ▶ Insert at *end* of list, using end pointer.
- ▶ For output, go through each bucket in order.

Running time: **$O(n + k)$**

Bucket Sort: Counting Sort++

Often want to sort *objects* based on *keys*:

- ▶ Each object has a key: integer in $\{0, 1, \dots, k - 1\}$
- ▶ **A** consists of **n** objects

Bucket Sort:

- ▶ Same idea as counting sort, but **B**[*i*] is bucket of objects with key *i*
- ▶ Bucket is a linked list with pointers to beginning and end
- ▶ Insert at *end* of list, using end pointer.
- ▶ For output, go through each bucket in order.

Running time: **$O(n + k)$**

Stable: if two objects have same key, order between them after sorting is same as before.

Radix Sort: Setup

What if k is much larger than n , e.g., $k = \Theta(n^2)$?

Radix Sort: Setup

What if k is much larger than n , e.g., $k = \Theta(n^2)$?

Radix sort: $O(n)$ time* for this case

Radix Sort: Setup

What if k is much larger than n , e.g., $k = \Theta(n^2)$?

Radix sort: $O(n)$ time* for this case

Setup:

- ▶ Numbers represented base 10 for historical reasons (all works fine in binary)
- ▶ Assume all numbers have exactly d digits (for simplicity)

Radix Sort: Setup

What if k is much larger than n , e.g., $k = \Theta(n^2)$?

Radix sort: $O(n)$ time* for this case

Setup:

- ▶ Numbers represented base 10 for historical reasons (all works fine in binary)
- ▶ Assume all numbers have exactly d digits (for simplicity)

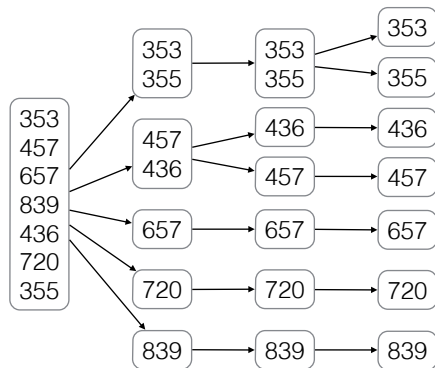
If you were sorting cards, with a number on each card, what might you do?

Radix Sort: Algorithm

Divide into **10** buckets by first digit, recurse on each bucket by second-digit, etc.

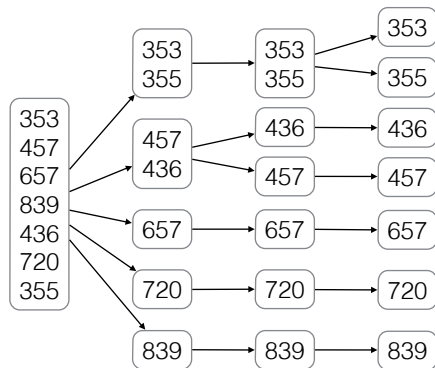
Radix Sort: Algorithm

Divide into **10** buckets by first digit, recurse on each bucket by second-digit, etc.



Radix Sort: Algorithm

Divide into **10** buckets by first digit, recurse on each bucket by second-digit, etc.



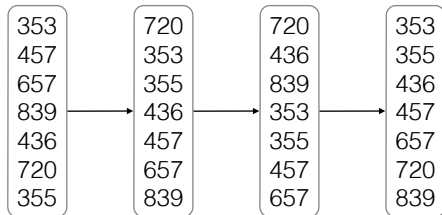
Works, but clunky

Radix-Sort: Algorithm (II)

More elegant (and surprising): one bucket, sorting from *least* significant digit to *most*!

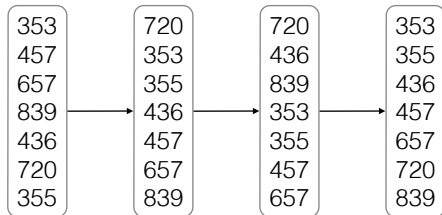
Radix-Sort: Algorithm (II)

More elegant (and surprising): one bucket, sorting from *least* significant digit to *most*!



Radix-Sort: Algorithm (II)

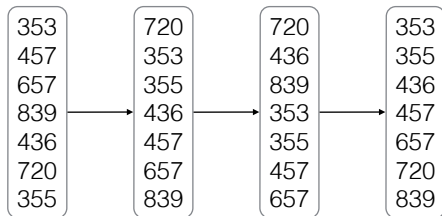
More elegant (and surprising): one bucket, sorting from *least* significant digit to *most*!



For iteration i , use bucket sort where key is i 'th digit and object is number.

Radix-Sort: Algorithm (II)

More elegant (and surprising): one bucket, sorting from *least* significant digit to *most*!



For iteration i , use bucket sort where key is i 'th digit and object is number.

Theorem

Radix sort from least significant to most significant is correct if the sort used on each digit is stable.

Least-Significant Radix Sort: Correctness

Proof.

Claim: After i 'th iteration, correctly sorted by last i digits (interpreted as # in $[0, 10^i - 1]$).

Least-Significant Radix Sort: Correctness

Proof.

Claim: After i 'th iteration, correctly sorted by last i digits (interpreted as # in $[0, 10^i - 1]$).
Induction on i .

Least-Significant Radix Sort: Correctness

Proof.

Claim: After i 'th iteration, correctly sorted by last i digits (interpreted as $\#$ in $[0, 10^i - 1]$).
Induction on i .

Base case: After first iteration, correctly sorted by last digit

Least-Significant Radix Sort: Correctness

Proof.

Claim: After i 'th iteration, correctly sorted by last i digits (interpreted as # in $[0, 10^i - 1]$).
Induction on i .

Base case: After first iteration, correctly sorted by last digit

Induction:

- ▶ Suppose correct for i
- ▶ After $i + 1$ sort:

Least-Significant Radix Sort: Correctness

Proof.

Claim: After i 'th iteration, correctly sorted by last i digits (interpreted as $\#$ in $[0, 10^i - 1]$).
Induction on i .

Base case: After first iteration, correctly sorted by last digit

Induction:

- ▶ Suppose correct for i
- ▶ After $i + 1$ sort:
 - ▶ If two numbers have different $i + 1$ digits, now correct.
 - ▶ If two numbers have the same $i + 1$ digit, were correct and still correct by stability.



Least-Significant Radix Sort: Running Time

Recall have n numbers, all numbers have d digits.

Least-Significant Radix Sort: Running Time

Recall have n numbers, all numbers have d digits.

bucket sorts:

Least-Significant Radix Sort: Running Time

Recall have n numbers, all numbers have d digits.

bucket sorts: d

Least-Significant Radix Sort: Running Time

Recall have n numbers, all numbers have d digits.

bucket sorts: d

Time per bucket sort:

Least-Significant Radix Sort: Running Time

Recall have n numbers, all numbers have d digits.

bucket sorts: d

Time per bucket sort: $O(n + k) = O(n + 10) = O(n)$.

Least-Significant Radix Sort: Running Time

Recall have n numbers, all numbers have d digits.

bucket sorts: d

Time per bucket sort: $O(n + k) = O(n + 10) = O(n)$.

Total time: $O(dn)$

Least-Significant Radix Sort: Running Time

Recall have n numbers, all numbers have d digits.

bucket sorts: d

Time per bucket sort: $O(n + k) = O(n + 10) = O(n)$.

Total time: $O(dn)$

Is this good? Bad? In between?

If all numbers distinct, $d \geq \log_{10} n \implies$ total time $O(n \log n)$

Least-Significant Radix Sort: Running Time

Recall have n numbers, all numbers have d digits.

bucket sorts: d

Time per bucket sort: $O(n + k) = O(n + 10) = O(n)$.

Total time: $O(dn)$

Is this good? Bad? In between?

If all numbers distinct, $d \geq \log_{10} n \implies$ total time $O(n \log n)$

Bad: not $O(n)$ time!

Good: "Size of input" is $N = nd$, so linear in size of input!

Least-Significant Radix Sort: Running Time

Recall have n numbers, all numbers have d digits.

bucket sorts: d

Time per bucket sort: $O(n + k) = O(n + 10) = O(n)$.

Total time: $O(dn)$

Is this good? Bad? In between?

If all numbers distinct, $d \geq \log_{10} n \implies$ total time $O(n \log n)$

Bad: not $O(n)$ time!

Good: “Size of input” is $N = nd$, so linear in size of input!

Improve to $O(n)$?

Fast Radix Sort

Change to go ***b*** digits at a time instead of just **1**.

- ▶ Kind of cheating: look at ***b*** digits in constant time.
- ▶ Necessary if we want time better than ***nd***

Fast Radix Sort

Change to go ***b*** digits at a time instead of just **1**.

- ▶ Kind of cheating: look at ***b*** digits in constant time.
- ▶ Necessary if we want time better than ***nd***

bucket sorts:

Fast Radix Sort

Change to go ***b*** digits at a time instead of just **1**.

- ▶ Kind of cheating: look at ***b*** digits in constant time.
- ▶ Necessary if we want time better than ***nd***

bucket sorts: ***d/b***

Fast Radix Sort

Change to go **b** digits at a time instead of just **1**.

- ▶ Kind of cheating: look at **b** digits in constant time.
- ▶ Necessary if we want time better than **nd**

bucket sorts: **d/b**

Time per bucket sort:

Fast Radix Sort

Change to go **b** digits at a time instead of just **1**.

- ▶ Kind of cheating: look at **b** digits in constant time.
- ▶ Necessary if we want time better than **nd**

bucket sorts: **d/b**

Time per bucket sort: **$O(n + k) = O(n + 10^b)$**

Fast Radix Sort

Change to go **b** digits at a time instead of just **1**.

- ▶ Kind of cheating: look at **b** digits in constant time.
- ▶ Necessary if we want time better than **nd**

bucket sorts: **d/b**

Time per bucket sort: **$O(n + k) = O(n + 10^b)$**

Total time: **$O\left(\frac{d}{b}(n + 10^b)\right)$**

Fast Radix Sort

Change to go ***b*** digits at a time instead of just **1**.

- ▶ Kind of cheating: look at ***b*** digits in constant time.
- ▶ Necessary if we want time better than ***nd***

bucket sorts: ***d/b***

Time per bucket sort: $O(n + k) = O(n + 10^b)$

Total time: $O\left(\frac{d}{b} (n + 10^b)\right)$

Set ***b*** = $\log_{10} n$. If ***d*** = $O(\log n)$, then time

$$O\left(\frac{d}{\log_{10} n} (n + n)\right) = O(n)$$

Fast Radix Sort

Change to go **b** digits at a time instead of just **1**.

- ▶ Kind of cheating: look at **b** digits in constant time.
- ▶ Necessary if we want time better than **nd**

bucket sorts: **d/b**

Time per bucket sort: **$O(n + k) = O(n + 10^b)$**

Total time: **$O\left(\frac{d}{b} (n + 10^b)\right)$**

Set **$b = \log_{10} n$** . If **$d = O(\log n)$** , then time

$$O\left(\frac{d}{\log_{10} n} (n + n)\right) = O(n)$$

Example: sorting integers between **0** and **n^{10}** . Then **d** should be about **$\log_{10} n^{10} = 10 \log_{10} n$** , as required.